Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

Mid-Semester Examination

October 2025

Max. Marks: 30 | **Duration: 1 Hrs**

Class: TE | **Semester: V**

Course Code: CE304/CS304 | **Program: CE/CSE**

Name of the Course: Theory of Computation

Instruction:

(1) All questions are compulsory

(2) Partial credit will be given, so show your work where appropriate

(3) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar, TM = Turing Machine.

(4) Show your reasoning wherever possible. No credit for merely stating the final answer even if correct.

Course Outcomes At the end of the course student will be able to:

(1) Design finite automaton for regular expressions and languages.

(2) Apply the properties of regular languages.

(3) Construct the grammar for a language and convert it into normal forms.

(4) Design and evaluate pushdown automata and Turing Machine for a language.

Q No.		Max. Marks	CO-BL
Q.1a	Construct a Deterministic Finite Automaton (DFA) over the alphabet $\Sigma = \{a, b\}$ that accepts all strings where the total number of a's is even and the string ends with ab. Demonstrate one example each of acceptance and rejection of strings by the DFA.	5	1-3
Q.1b	Construct an ϵ–NFA over $\Sigma = \{0, 1\}$ that accepts the language: L = { strings that contain 101 as a substring } and convert it to an NFA without epsilon.	5	1-3
Q.2a	State True or False with appropriate justification. Let L be the set of strings in the intersection of $(ab)^*(ba)^*$ with $(ba)^*(ab)^*$. Then $L = \{\epsilon\}$.	5	2-3
Q.2b	Let $L = \{a^n b^n \mid n > 100\}$. Is the following language regular? Explain why/why not.	5	2-3
Q.3a	Construct the regular expression for DFA with binary strings that does not contain 11 as substring using Arden's theorem.	5	2-3
Q.3b	Construct a Moore machine take all the strings of a's and b's as input and counts the no of occurrences of substring "ab" in terms of number of 1's, over the input $\Sigma = \{a, b\}$, output $\Delta = \{0, 1\}$.	5	1-3